

ReactJS

Software System Development



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Manu Awasthi

@mnwsth · 19h

Graduating from a PhD program in 4 years



9



11



304



Palki Sharma liked



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Am I the only one who misses @palkisu on @WIONews these days? When is she coming back on Gravitas Plus?



23



28



462



Sai Anirudh K
@SaiAnirudhK1

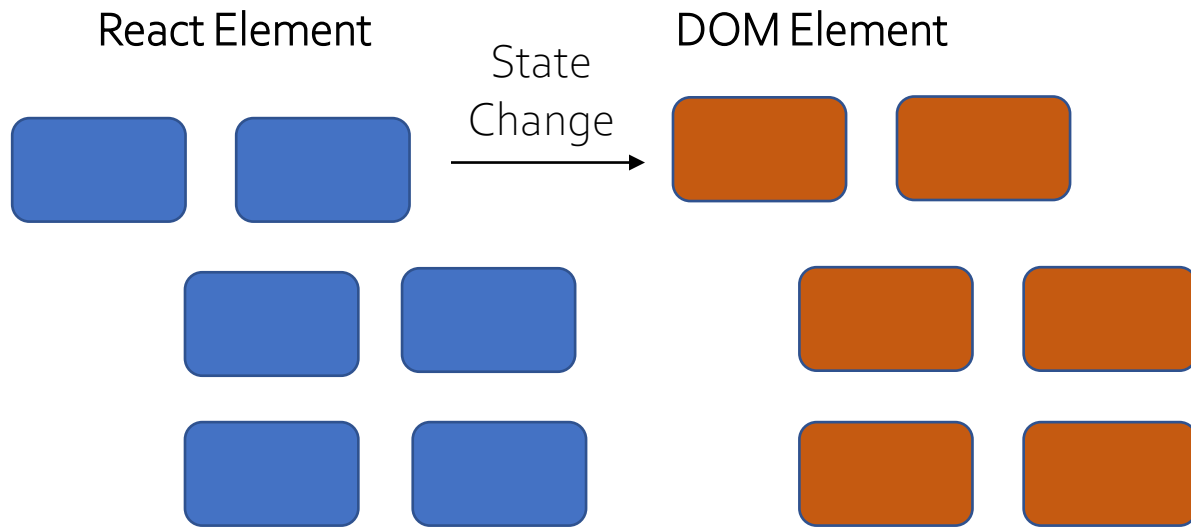


Deepak Sen

```

Class Tweet {
  state = {};
  render() {
    }
  }
  
```

→ React Element



```

const element = document.querySelector();
element.classList.add();
element.addEventListener ();
//React to update the State Change of DOM rather updating DOM
  
```



Node package manager



```
~sudo npm -i -g create-react-app@1.5.2
```

Visual Studio Code Plugin

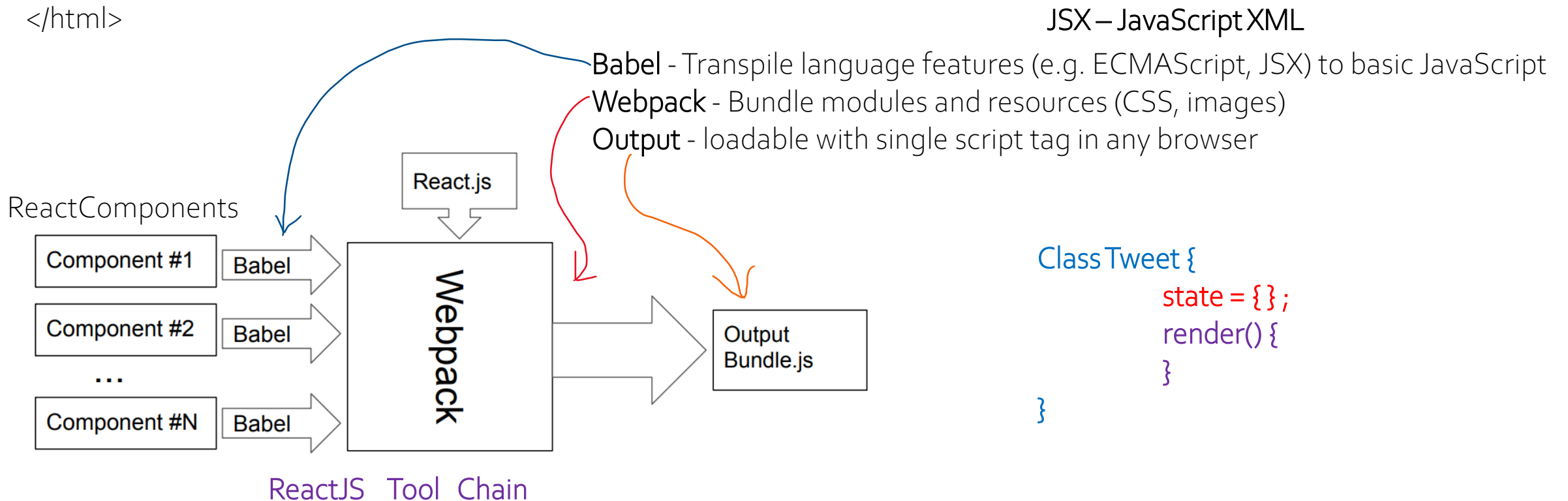
ReactSimpleSnippets – Bruke Holland
 Prettier – Code formatter – Esben Peterson

```

<!doctype html>
<html>
  <head>
    <title>SSD Example</title>
  </head>
  <body>
    <div id="reactapp"></div>
    <script src="./webpackOutput/reactApp.bundle.js"></script>
  </body>
</html>

```

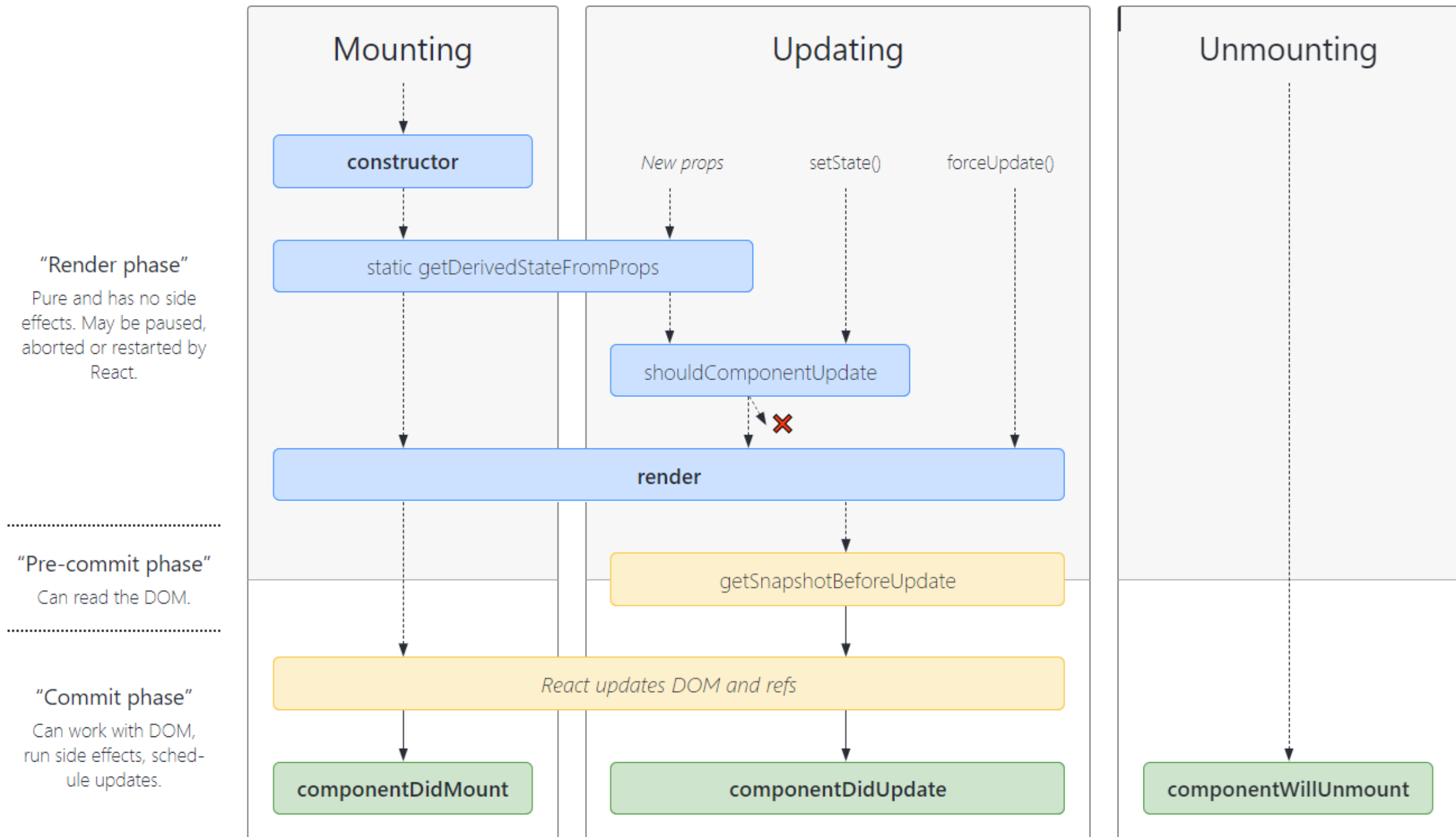
ReactJS applications come as a JavaScript blob that will use the DOM interface to write the view into the div.



```

Class Tweet {
  state = {};
  render() {
  }
}

```



Component

Components are self-contained reusable building blocks of web application.

React components are basically just idempotent functions (same input produces same output) i.e. $F(f(x)) = f(x)$

They describe your UI at any point in time, just like a server-rendered app.

Created using `React.createClass()`
The only required method is `render()`
Inserted into DOM
using `React.renderComponent()`

```
var React = require('react'),
    SimpleView = React.createClass({
      render: function () {
        return <h1><strong>Example
1:</strong> A simple component</h1>;
      }
    });

React.renderComponent(SimpleView(
),
document.getElementById('example'))
;
```

Props

Passed down to component from parent component and represents data for the component. It is accessed via **this.props**

```
render: function() {
```

```
  var someProp = 'bar';
```

```
  console.log('component render()', this.props);
```

```
  return <div>
```

```
    <AnotherComponent foo={someProp} model={this.props.model} />
```

```
  </div>;
```

```
}
```

State

Represents internal state of the component.

Accessed via **this.state**

When a component's state data changes, the rendered markup will be updated by re-invoking `render()` method

```
render: function() {  
  return <h3>Click count:  
    <span className='label label-default'>{this.state.clicks}</span>  
  </h3>;  
}
```

JSX

XML-like syntax for
generating
component's HTML

Easier to read and
understand large
DOM trees

Translates to plain
JavaScript using
react-tools

```
/** @jsx React.DOM */
```

```
render: function () {  
  return <div>  
    <h2>  
      <strong>Example 4:</strong> React App  
    </h2>  
  </div>;  
}
```

```
/** regular DOM */
```

```
render: function () {  
  return React.DOM.div(null,  
    React.DOM.h2(null, React.DOM.strong(null, "Example 4:"), " React App")  
  );  
}
```

Virtual DOM

- The **virtual DOM** is used for efficient re-rendering of the DOM
- React aims to re-render the virtual tree only when the state changes
- Uses 2 virtual trees (new and previous) to find differences and batch update real DOM
- Observes data changes (setState) and does dirty-checking to know when to re-render component
- Whenever possible, does not update entire component in real DOM
 - only computes a patch operation that updates part of the DOM